

SUMMARY TABLE: BIOLOGY GRADE 10

Sub-Strand 3.3: Animal Gaseous Exchange and Respiration — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.3: Animal Gaseous Exchange and Respiration
Driving Question	DRIVING QUESTION: Why does my body work so much harder to get oxygen into my cells during exercise — and what happens inside those cells when there is not enough?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Your Body Changes When You Move — Anchoring the Phenomenon	Learners measured their own resting breathing rate and pulse rate, performed 2 minutes of step-up exercise, then measured both rates again immediately after. The class data table showed that breathing rate approximately doubled (e.g., 14 → 28 breaths/min) and pulse rate rose sharply (e.g., 72 → 120+ bpm) in every learner. Some learners also reported a burning sensation in their thigh muscles during or after the exercise.	Learners confirmed through their own body data that both breathing rate and pulse rate increase measurably and universally during vigorous physical exercise. They observed that the two systems — breathing and heartbeat — appear to respond together at the same time. They also noted that the burning sensation in leg muscles is a separate, puzzling observation that their current knowledge cannot yet explain.	Learners generated their own first-draft explanations in small groups: muscles need more energy during exercise, oxygen is somehow needed for that energy, and the lungs and heart must be working together to supply it. These explanations are partial and incomplete — they do not yet account for how oxygen moves from the lungs to the muscles, what happens to oxygen inside cells, or why muscles burn even when oxygen is being breathed in. These gaps are now posted on the Driving Question Board and will be systematically answered across the unit.
Lesson 2 Characteristics of Respiratory Surfaces — What Makes a Good Gas Exchange Surface?	Learners examined diagrams and slides of four respiratory surfaces — earthworm skin, fish gill lamellae, insect tracheal tubes, and mammalian alveoli — and completed a comparison table recording surface area, wall thickness, moisture, and blood/fluid supply for each. They also observed limewater turning milky when exhaled air was blown through it.	All efficient respiratory surfaces share four structural features: large surface area, thin walls (one to a few cells thick), a moist surface (to keep gases in solution for diffusion), and a rich blood or fluid supply (to maintain a concentration gradient). Mammalian alveoli have the most extensive surface area and the thinnest walls, making them highly adapted for the rapid, high-volume gas exchange needed during exercise. Carbon dioxide is a real,	The structural features of the alveolus directly explain why the breathing rate can double during exercise and still supply enough oxygen: the enormous combined surface area of millions of alveoli maximises the number of gas molecules that can diffuse simultaneously, while the one-cell-thick moist wall minimises the distance gases must travel. This structural efficiency is the reason a human sprinting near Uhuru Park in Nairobi can sustain high muscle

		chemically detectable product that exits through the respiratory surface.	activity — the respiratory surface is built for exactly this demand.
Lesson 3 Structure and Adaptations of Respiratory Surfaces	Learners observed diagrams and/or specimens of three respiratory surfaces — human alveoli, tilapia gill lamellae (Lake Victoria context), and insect tracheoles — at structured observation stations. They measured estimated wall thicknesses, counted lamellae and capillaries, and traced the branching path of tracheoles to muscle cells.	Learners learned that all efficient respiratory surfaces share four key structural adaptations: (1) large surface area, (2) thin walls for a short diffusion distance, (3) a moist surface so oxygen can dissolve before crossing, and (4) a rich blood supply to maintain a steep concentration gradient. These adaptations explain why the alveolus can double its oxygen-delivery rate within seconds when exercise begins.	Learners explained how the four adaptations of the alveolus solve the problem of getting oxygen into the blood fast enough to meet the doubled demand during step-ups. They also explained why a tilapia's gill lamellae must have the same adaptations in an aquatic environment, and began to model the pathway: oxygen → alveolus → blood → (next question) → muscle cell.
Lesson 4 Mechanisms of Gaseous Exchange in Humans — External Respiration	Learners watched an ARES offline animation showing diaphragm and intercostal muscle movements, thoracic volume changes, and pressure gradients during inhalation and exhalation. Learners also constructed and manipulated an improvised bell-jar diaphragm model (plastic bottle, balloons, Y-straw) and directly observed inner balloons (lungs) inflate when the base balloon (diaphragm) was pulled downward.	Learners learned that breathing is a mechanical, pressure-driven process: during inhalation, the diaphragm contracts and flattens while external intercostal muscles raise the rib cage, increasing thoracic volume and decreasing intrathoracic pressure below atmospheric pressure, causing air to flow in; during exhalation, these muscles relax, thoracic volume decreases, pressure rises, and air is expelled. Key vocabulary consolidated: tidal volume, inspiration, expiration, pleural cavity.	Learners used the Claim–Evidence–Reasoning framework to explain that the doubled breathing rate observed after the step-up exercise is achieved by the diaphragm and intercostal muscles contracting more frequently and more forcefully, creating larger and more rapid pressure gradients so that more air — and therefore more oxygen — is moved to the lungs per minute. They connected the physical model behaviour to their own body data from the anchoring phenomenon.
Lesson 5 Mechanisms of Gaseous Exchange in Humans — Alveolar & Cellular Exchange	Learners examined class data showing that both breathing rate and heart rate rose sharply after 2 minutes of step-ups (e.g., pulse from 72 to over 120 bpm). They read evidence on the ARES platform describing how oxygen diffuses across alveolar walls into capillaries, binds to haemoglobin in red blood cells, is pumped by the heart through the circulatory system, and diffuses from muscle capillaries into muscle cells down a concentration gradient — while CO ₂ travels the reverse route.	Oxygen does not travel directly from the lungs to the muscles — it is carried by haemoglobin in red blood cells through the circulatory system. The heart acts as a pump that drives oxygenated blood from the lungs to every cell in the body. During exercise, the heart beats faster to increase the rate of oxygen delivery to muscle cells and the rate of carbon dioxide removal. Diffusion at both the alveolar surface and the muscle capillary surface is driven by concentration gradients: oxygen moves from high concentration to low, and carbon dioxide moves in the opposite direction.	The complete journey of one oxygen molecule — alveolus → capillary wall (diffusion) → haemoglobin in red blood cell → pulmonary vein → left side of heart → aorta → muscle capillary → muscle cell (diffusion) — explains why both breathing rate AND heart rate rose during the step-up activity. The lungs increase oxygen supply; the heart increases delivery speed. Together, these two systems ensure muscle cells receive enough oxygen to keep releasing energy during exercise. This closes the puzzle of the heart–lung connection first raised in Lesson 1.
Lesson 6 Aerobic Respiration: Unlocking Energy from Ugali Inside Your Cells	Learners observed the word equation (glucose + oxygen → carbon dioxide + water + ATP) and the balanced chemical equation ($C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + ATP$) for aerobic respiration, framed	Learners learned that aerobic respiration occurs in the mitochondria of cells and uses glucose (from digested food such as ugali) and oxygen (delivered by the breathing and circulatory systems investigated in Lessons	The lesson explained the core mechanism behind the anchoring phenomenon: when learners performed step-ups, their muscle cells needed more ATP to power contractions, so aerobic respiration in the

	through the context of a Rift Valley maize farmer fuelled by ugali. They observed a printed mitochondrion diagram and annotated the entry of reactants and exit of products.	1–5) to produce ATP (usable energy for muscle contraction), carbon dioxide, and water. The increased demand for ATP during exercise explains why both breathing rate and pulse rate rise — the body must deliver more oxygen and glucose to the mitochondria and remove more carbon dioxide.	mitochondria increased its rate, consuming more oxygen and glucose and producing more carbon dioxide — which is precisely why breathing rate doubled and pulse rate spiked. The equation now gives a molecular explanation for every whole-body change measured on Day 1.
Lesson 7 Anaerobic Respiration: Yeast Balloons, Burning Legs, and Oxygen Debt	Learners observed a balloon inflate over a bottle containing yeast, glucose, and warm water over 20 minutes, recording increasing balloon circumference at 5-minute intervals. They also observed a video clip of a Kenyan distance runner experiencing visible muscle fatigue at the finish line.	Learners learned that anaerobic respiration in yeast produces ethanol and carbon dioxide (inflating the balloon) and that anaerobic respiration in animal muscle cells produces lactic acid. They learned that both pathways release energy without oxygen, but release far less ATP than aerobic respiration. They defined oxygen debt as the extra oxygen required after intense exercise to break down accumulated lactic acid back to CO ₂ and water.	Learners explained in writing why lactic acid builds up in leg muscles during intense step-up exercise — because oxygen supply to muscle cells is insufficient to sustain aerobic respiration alone, forcing cells to switch to the lactic acid pathway. They connected the balloon evidence (CO ₂ production without oxygen) to the broader principle of anaerobic energy release, resolving the 'burning legs' mystery that has been on the Driving Question Board since Lesson 1.
Lesson 8 Respiratory Quotient (RQ): What Fuel Is Your Body Burning?	During the step-up exercise at the start of our unit, we observed that breathing rate doubled and pulse rate rose sharply — and some learners felt a burning sensation in their leg muscles even while breathing hard.	Today we learned that the body can burn different fuels — maize starch (carbohydrate) from ugali, fat from nyama choma, and protein from githeri — and each fuel produces a different ratio of CO ₂ to O ₂ , measured as the Respiratory Quotient (RQ). Carbohydrate gives RQ = 1.0, fat gives RQ ≈ 0.7, and protein gives RQ ≈ 0.9. During prolonged exercise, the body shifts from burning carbohydrate to burning fat, which requires MORE oxygen — explaining why the lungs and heart must work even harder.	The burning sensation in leg muscles during the step-up exercise is connected to a shift in fuel source: as exercise continues, muscles switch toward fat (lower RQ), demanding more oxygen per unit of energy. This is why breathing rate and heart rate rise so sharply — the body is responding not just to increased energy demand, but to a change in the type of fuel being burned inside the muscle cells.
Lesson 9 Practical & Data Analysis: Factors Affecting the Rate of Respiration	Yeast–sugar suspension in water baths at 10 °C, 25 °C, 37 °C, and 50 °C produced different rates of CO ₂ bubbles per minute. The highest bubble rate was recorded at 37 °C; the rate at 50 °C was lower than at 37 °C, and the rate at 10 °C was the lowest of all.	The rate of cellular respiration is controlled by enzymes. Enzymes work faster as temperature rises — up to an optimum of approximately 37 °C for yeast and human cells. Above the optimum, the enzyme's shape is permanently altered (denaturation) and the reaction rate drops sharply. Cellular respiration also releases heat as a by-product, which is why body temperature rises during exercise.	After the step-ups in Lesson 1, learners felt their muscles burn and their skin become warm. We can now explain this: the enzymes driving cellular respiration in muscle cells released metabolic heat as respiration rates surged to meet energy demand. The body must maintain temperature close to 37 °C — the optimum for its enzymes — which is why sweating and increased blood flow to the skin begin during vigorous exercise in the Kenyan heat.

Lesson 10 Factors Affecting Rate of Respiration (continued) & Importance of Gaseous Exchange	Learners analysed two graphs: (1) glucose concentration vs. CO₂ output in yeast, showing respiration rate rises then plateaus; (2) aerobic respiration rate in Kenyan runners tested at Nairobi, Eldoret, and a high-altitude Mt. Kenya training camp, showing rate decreases with altitude despite identical workload. Learners also annotated a text box on surface area and gaseous exchange, connecting alveolar surface area (~70 m²) to diffusion rate via Fick's Law.	Three additional factors affect the rate of cellular respiration: (1) Substrate concentration — increasing glucose increases respiration rate up to the point of enzyme saturation, after which rate plateaus; (2) Oxygen availability — reduced O₂ (e.g., at high altitude in the Rift Valley highlands) slows aerobic respiration, forcing cells into anaerobic respiration and producing lactic acid — the source of the burning sensation felt during step-ups in Lesson 1; (3) Surface area of the respiratory membrane — a larger surface area (obeying Fick's Law) allows greater O₂ diffusion per unit time, raising the ceiling on aerobic respiration rate. Respiration is important for: movement (e.g., Kipchoge's marathon performance), growth, active transport (e.g., glucose absorption from githeri), heat regulation (e.g., working in cold Kericho highlands), and reproduction.	The doubled breathing rate and sharply raised pulse rate observed after step-ups in Lesson 1 reflect the body's urgent attempt to increase oxygen delivery to muscle cells so they can maintain aerobic respiration at the rate demanded by exercise. When oxygen delivery cannot keep pace — as it cannot at high altitude or during very intense exercise — cells switch to anaerobic respiration, producing lactic acid and causing the burning sensation many learners felt. The efficiency of this entire system depends on substrate availability, oxygen availability, and the surface area of the respiratory membranes — all three factors that were investigated in this lesson.
Lesson 11 Model Building and Driving Question Board Consolidation	Learners observed an incomplete exemplar cumulative model and identified missing components including the anaerobic lactic acid pathway, comparative respiratory surfaces panel, breathing mechanics pressure labels, haemoglobin as the oxygen carrier, and factors affecting respiration rate. Learners also observed their own model improve in response to specific peer feedback written using an explicit checklist.	Learners synthesised knowledge from all ten prior lessons into a single, five-zone annotated cumulative model covering comparative respiratory surfaces, human breathing mechanics, the full gas exchange and transport pathway, aerobic and anaerobic respiration equations with RQ, and factors affecting respiration rate. Learners confirmed that every question on the Driving Question Board — all originally triggered by the Lesson 1 step-up phenomenon — can now be answered with biological evidence.	Learners explained in a written model caption how the doubling of breathing rate and sharp rise in pulse rate during step-ups results from the integrated action of alveolar gas exchange, haemoglobin-mediated oxygen transport, aerobic cellular respiration in muscle mitochondria, and — when oxygen supply is insufficient — the anaerobic production of lactic acid that causes the burning sensation felt in Lesson 1. This explanation directly advances the driving question by demonstrating that the body works harder to supply oxygen because mitochondria consuming glucose in aerobic respiration continuously deplete local oxygen, and that when even doubled breathing cannot meet demand, anaerobic respiration fills the energy gap at the cost of lactic acid accumulation.
Lesson 12 Summative Review & Transfer: Explaining the Body's Response to Exercise	Learners revisited their own Lesson 1 resting and post-exercise breathing rate and pulse rate data, and compared their initial naive model of exercise physiology with their final integrated model built across 12 lessons.	Learners integrated the complete explanatory chain: muscle ATP demand → aerobic cellular respiration → CO₂ build-up → increased breathing rate and depth → alveolar gas exchange → haemoglobin transport → aerobic/anaerobic pathways in	Learners can now fully explain the anchoring phenomenon: why breathing rate doubles during exercise (increased CO₂ and decreased O₂ in blood stimulate faster, deeper breathing to meet muscle cell demand), why the heart beats faster

		muscle cells → lactic acid accumulation → oxygen debt. They applied this framework to the novel scenario of Akinyi swimming the Winam Gulf of Lake Victoria.	simultaneously (to pump oxygenated blood more rapidly to working muscles), and why legs burn even during hard breathing (when oxygen delivery cannot meet demand, muscle cells switch to anaerobic respiration producing lactic acid, which is acidic and causes the burning sensation). They can also explain why heavy breathing continues after exercise stops — to repay the oxygen debt by oxidising accumulated lactic acid.
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